

Problem Set 1
Intermediate Microeconomics
Spring 2023

1. Chapter 7's Exercises 2.12, 3.2
2. (Opportunity Cost) Jenny is considering starting a new business selling organic groceries. For one year, it would cost her \$350,000 rent store space and buy the groceries from a wholesaler, and she would have to quit her current job and give up \$70,000 in salary. She would also need to spend \$100,000 of her own savings on cash registers (money that could be earning 8% interest) but she could re-sell the cash registers at the end of the year for \$100,000. Assume there are no other expenses.
 - a. What is Jenny's economic cost of starting and running the business for a year?
 - b. If her grocery store is expected to bring in an accounting profit of \$75,000 in one year, should she start the business? Why or why not?
3. (Cost Functions) Suppose that a firm's production function is given by

$$Q = \sqrt{L} + \sqrt{K} + \sqrt{M}.$$

The input prices of labor, capital, and materials are $w=1$, $r=1$, and $m=1$, respectively.

- a. Given that the firm wants to produce 12 units of output, what is the solution to the firm's long-run cost minimization problem?
- b. Given that the firm wants to produce 12 units of output, what is the solution to the firm's short-run cost minimization problem when $K=4$?
- c. Given that the firm wants to produce 12 units of output, what is the solution to the firm's short-run cost minimization problem when $K=4$ and $L=9$?

2.12

$$C(q) = 0.55q^{1.67} + \frac{800}{q}$$

$$AVC(q) = 0.55q^{0.67} + \frac{800}{q^2}$$

$$\frac{dAVC(q)}{dq} = 0.55 \times 0.67 q^{-0.33} - \frac{1600}{q^3}$$

$$\frac{dAVC(q)}{dq} \Rightarrow q^{2.67} = 4347.826$$

$$q = 23$$

$$C(q) = 0.55q^{1.67} + \frac{800}{q} + 400$$

$$AC(q) = 0.55q^{0.67} + \frac{800}{q^2} + \frac{400}{q}$$

3.2 $r = 1,000$ $w = 200$

$$MP_K = 200 \quad MP_L = 50$$

$$\frac{MP_L}{MP_K} = \frac{w}{r} \rightarrow \frac{1}{4} \frac{50}{200}, \frac{200}{1000} \frac{1}{5}$$

Cost minimum

$$MRTS > \frac{w}{r} \Rightarrow$$

It is not minimizing cost

It should hire more workers or use less capital.



2. 350,000 rent store

70,000 give salary

100,000 cash registers 8% interest
8,000

$$EC = 350,000 + 70,000 + 8,000 + 100,000$$

$$\text{Accounting profit} = 15,000 \Rightarrow 15,000 < 18,000$$

$$70,000 + 8,000 = 18,000 \rightarrow \text{Not to start the business}$$

$$3. Q = \sqrt{L} + \sqrt{K} + \sqrt{M}$$

$$w=1 \quad r=1 \quad m=1$$

$$a. 12 = \sqrt{L} + \sqrt{K} + \sqrt{M} \quad \frac{\frac{1}{2} L^{-\frac{1}{2}}}{\frac{1}{2} K^{-\frac{1}{2}}}$$

$$\frac{MP_L}{MP_K} = \frac{1}{1} \Rightarrow K=L \quad \left(\frac{K}{L}\right)^{\frac{1}{2}} = 1$$

$$\frac{MP_L}{MP_M} = \frac{1}{1} \Rightarrow M=L$$

$$\Rightarrow L=M=K$$

$$12 = 3\sqrt{L}$$

$$4 = \sqrt{L} \Rightarrow L=16=M=K$$

$$16 \times 3 = 48$$

$$b. \quad q = \sqrt{L} + \sqrt{K} + \sqrt{M}$$

$$q = \sqrt{L} + 2 + \sqrt{M}$$

$$10 = \sqrt{L} + \sqrt{M}$$

$$\frac{MP_L}{MP_M} = \frac{r}{m} = 1 \Rightarrow L = M$$

$$25 \times 2 + 4 = 54$$

$$5 = \sqrt{L} \quad L = 25 = M$$

$$c. \quad q = \sqrt{L} + \sqrt{K} + \sqrt{M}$$

$$12 = \sqrt{9} + \sqrt{4} + \sqrt{M}$$

$$9 + 4 + 49 = 62$$

$$12 = 3 + 2 + \sqrt{M}$$

$$7 = \sqrt{M} \Rightarrow M = 49$$

$$1.3 \text{ Oppo cost} = 9$$

$$\text{Sunk cost} = 1$$

$$3.1 \quad q = L + K$$

$$MRTS = \frac{MP_L}{MP_K} = \frac{w}{r}$$

$$C = wL + rK$$

$$\text{if } w > r \Rightarrow C = rK$$

$$\text{if } w < r \Rightarrow C = wL$$

$$w = r \Rightarrow C = wL + rK \text{ at any quantity.}$$